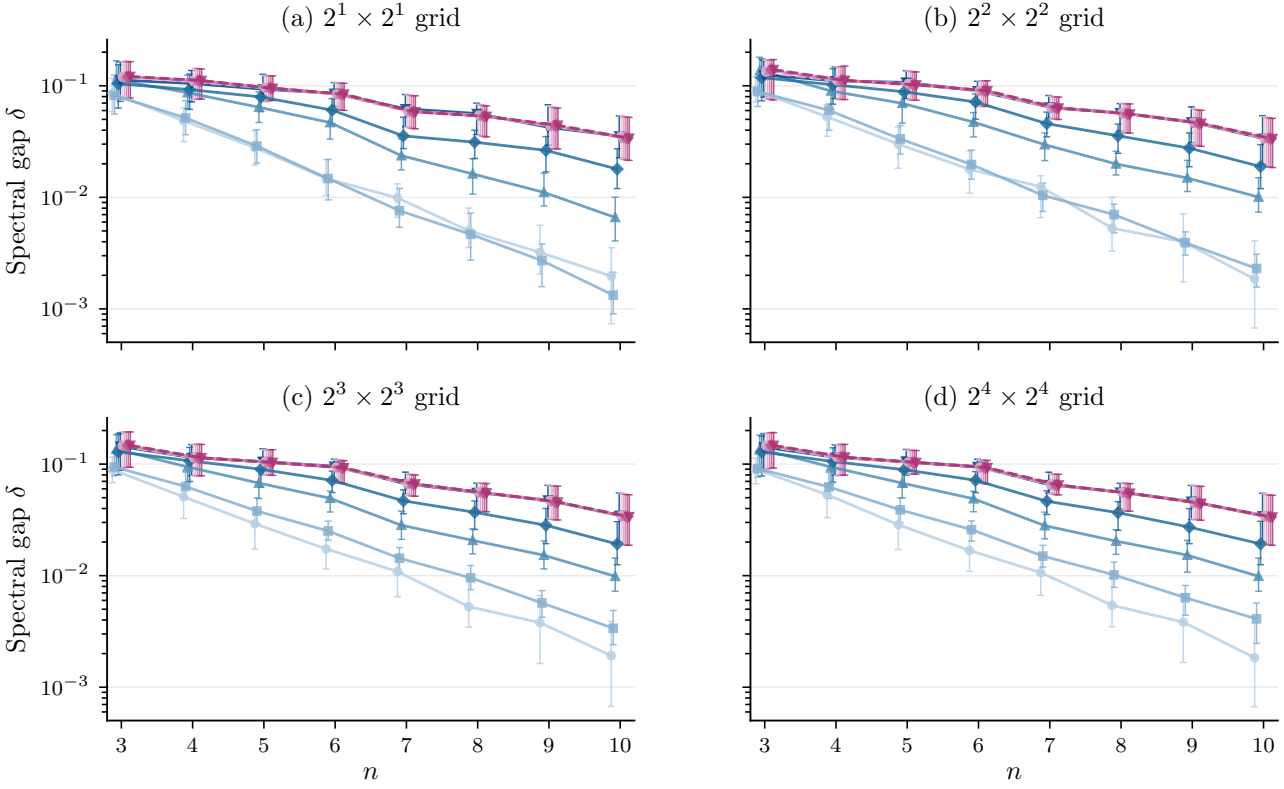

(a) $2^1 \times 2^1$ grid
(b) $2^2 \times 2^2$ grid
(c) $2^3 \times 2^3$ grid
(d) $2^4 \times 2^4$ grid
Spectral gap δ
10^{-1}
10^{-2}
10^{-3}
3
4
5
6
7
8
9
10
n
$r = 1$
$r = 2$
$r = 5$
$r = 10$
$r = 20$
$r = 30$
$r = 40$
$r = 50$
$r = 75$
$r = 100$